Time-Attraction Law v16.0 FINAL
Correcting Newton's and Hubble's Laws via Density-Dependent Time Coherence

David Unbekannte -- 2026-10-08 -- Concept DOI: 10.5281/zenodo.22983681 -- Previous Record DOI: 10.5281/zenodo.23220318 (v15, attached file was v14.1) -> NEW v16.0
GitHub: github.com/unbekanntdavid-glitch/Time-Attraction
Constants: C0=5, rho_c=1e-23 kg/m3, L0=1 kpc=3.085677581e19 m, hz=0.2 kpc, Ydisk=0.5, Ybulge=0.7 at 3.6 um, G=6.67430e-11 SI, lambda0=1.3e26 m, M_sun=1.98847e30 kg, kpc=3.085677581e19 m

1. Pair functional and effective density (v16 fix for mechanics audit)

The audit found non-conservative force and action-reaction violation if rho_eff is defined at test point only. v16 defines symmetric pair functional:

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q(x1,x2) = [rho_local(x1)+rho_local(x2)] / (2 rho_c) + (1/(L0 rho_c)) * Integral_{x1}^{x2} rho_b(l) dl
(Eq 1)
rho_eff = rho_c * q , C(r) = 1 + C0 * exp(-q) = 1+5*exp(-q) , Phi_t(r)=1/C(r) (Eq 2)
Integral is along straight source-test segment length r=|x2-x1|, rho_b is rest-frame baryonic density
excluding self-fields, rho_local(x) is 3D top-hat average over radius 0.3 kpc centered at x. This makes
q_forward = q_reverse. Observer sightlines and galaxy-center rays are distinct and not used for pair law.
For constant rho_b=rho0: q=(rho0/rho_c)*(1+r/L0) and C' = -5*(rho0/(rho_c*L0))*exp(-q) !=0, so even
uniform density gives r-dependence because of integrated term.
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2. Force and conservative potential (fixes Fp vs U0 mismatch)

Printed radial force on body 2: Fp,2 = -G m1 m2 C(r) n / r^2 (Eq 3) with n = (x2-x1)/r, A=G m1 m2. Potential U0 = -A C(r)/r gives F0 = -Grad2 U0 = -A C n / r^2 + (A/r) Grad2 C, so F0-Fp = (A/r) Grad2 C. Equality requires Grad2 C=0. For radial C(r): F0,r = -A[C/r^2 - C'/r], Fp,r = -A C/r^2. With C'<0, U0 more attractive. The conservative potential generating Fp as central force with U(inf)=0 is:

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Ur(r) = -A * Integral_r^inf C(s)/s^2 ds , -dUr/dr = -A C(r)/r^2 = Fp,r (Eq 4)
For finite bounded C, quadrature converges for r>0. U0 reproduces Fp only when C'=0.
Outside r=0: curl2 Fp,2 = -A (Grad2 C cross n)/r^2. Curl-free iff C has no angular gradient on spheres
about source. Meridional loop r1->r2 at theta1, then arc at r2, back at theta2: Wp/A =
-Integral_{r1}^{r2}[C(r,theta1)-C(r,theta2)] dr/r^2. For U0 transverse (A/r)GradC cancels loop work.
Disk regularized with eps=0.03 kpc: rho_b=0.3 rho_c exp(-R/2kpc) exp{-[sqrt(z^2+eps^2)-eps]/0.3kpc}.
Then off-plane Fp has nonzero curl, e.g. loop 1->2 kpc, theta1=0.15, theta2=0.8 rad: Wp/A=+0.405424
kpc^-1 (audit reproduced). In midplane angular gradient zero.
Action-reaction: old single-point rho_local(x2) gives q_forward=0.279236 vs q_reverse=0.156652 in disk
example. With symmetric Eq1 evaluate same q for both and impose F1=-F2 algebraically. Then instantaneous
action-reaction holds even with environmental dependence, but momentum exchange with medium requires its
dynamics. Translating both endpoints changes environment, so F1+F2=(A/r)(Grad1+Grad2)C generally !=0 if
rho_b is fixed background.
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Sufficient conditions for conservative central pair (v16 declaration): (i) C=f(r) single-valued differentiable depends only on separation, no orientation/absolute position/observer; (ii) Interaction Ur as Eq4 with finite reference; (iii) Same pair rule for both bodies. Spherical density centered on fixed source plus fixed radial prescription sufficient for test-body central force, not by itself translationally invariant two-body theory.

3. Operational density for SPARC (fixes 6160x mismatch and C=1 everywhere)

Public SPARC inputs Rotmod_LTG.zip 175 files SHA256 0a80cc90714828cc28b7dd57923576714d209f2490328c087c4a4ad607faf588 do NOT directly give resolved 3D rho_local. Literal printed rho ~ v_bar^2/(G r hz) is baryons-only:

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At NGC3198 r=30.08 kpc Vobs=146+-2 km/s, Vgas=45.29, Vdisk=76.07, Vbul=0, Ydisk=0.5: v_bar^2=Vgas^2+0.5
Vdisk^2=4944.5 (km/s)^2, v_bar=70.317 km/s, C_required=(146/70.317)^2=4.311. But rho_print=
v_bar^2*1e6/(G r_m hz,m)=1.293e-20 kg/m3=1293.3 rho_c, C~1, Vpred~70.3 km/s, residual +75.683 km/s
(+37.8 sigma). PDF example instead lists rho_eff=2.1e-24, 6160x smaller.
v16 fix replaces with photometric proxy, explicitly assumption:

rho_disk(R,0)= Ydisk * SBdisk(R) * M_sun/pc^2 / (2 hz), assume exp(-|z|/hz) (Eq5)
Sigma_gas(r)=1.33 M_HI exp(-r/Rg)/(2 pi Rg^2), Rg=2 R_disk, same hz (Eq6)
rho_bulge proxy = 3 Ybulge Vbul^2*1e6/(4 pi G r_m^3) (spherical enclosed-mass proxy)
rho_local = 3D top-hat 0.3 kpc average of rho_disk+rho_gas+rho_bulge, rho_integrated = (1/L0)
Integral_0^r rho_proxy(s) ds along midplane radial path centre->r, linearly interpolated between
measured radii, held constant to innermost radius. This yields rho~1e-24 at 30 kpc allowing C>1. Force
multiplier applied as V_pred^2 = C v_bar^2 per p.2, acknowledging that spatially varying nonlocal force
cannot be represented by multiplying existing disk curve by one local C; tests are of speed prescription.
Sample cuts: Q in {1,2}, Inc>30deg before residuals; Q=3 removes 12, Inc<=30 removes 14 further, leaving
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149/175 galaxies, 3137 points, all finite r>0 errV>0. Vgas sign: SPARC has 341 negative Vgas entries
(outward gas). Squared as printed loses sign; sensitivity check sign(Vgas)*Vgas^2 leaves 3135 points.
Pooled chi2: baryons-only C=1 => 621.92/3137, median galaxy chi2/n_g=118.46, RMSE 59.93 km/s, 0 points
C>1.01; literal printed same; spherical mean local-only 546.68/112.65/56.01/526 points.
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4. Redshift and distance (fixes inconsistent z prescriptions)

v14.1 had three inconsistent laws: (i) ODE dPhi_t/dl = -Phi_t rho_eff/(rho_c L0) => Phi_t=Phi_t(0)exp[-Integral rho dl/(rho_c L0)], (ii) Phi_t=1/[1+5 exp(-rho_eff/rho_c)], (iii) 1+z=1/Phi_t giving z=5 at zero density and zero distance, vs z=Integral (1-Phi_t)dl/lambda0 giving z=0 at d=0. ODE solution is exponential of optical-depth integral, not algebraic.
v16 fixes to ONE consistent law (path integral primary):

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Phi_t(l)=1/C(l)=1/[1+5 exp(-q(l))] (Eq7)
z(d)= Integral_0^d [1-Phi_t(l)] dl / lambda0 , lambda0=1.3e26 m ~ c/H0 dimensionless (Eq8)
DL = (1+z) * Integral_0^d dl/Phi_t(l) (Eq9)
1+z=1/Phi_t is only local constant-Phi_t approximation, not general. For d=0, z=0. Background H(a) not
derived from rho(a) here; luminosity distance alone does not fix angular size without optical geometry
and distance-duality rule. For BAO we must supply: (1) z(chi) from Eq8, (2) spatial/optical metric
specifying transverse DM(z) and DA=DM/(1+z) with duality, (3) radial DH(z)=d chi/dz, (4) calibrated
ruler rd. Gravitational multiplier alone fixes none, hence v15 had tal_prediction_available=False. v16
declares Eq8-9 as definition, but full DM/DH/rd mapping still requires explicit metric, so BAO test
remains conditional.
Printed Gaussian bump R=1+0.03 exp[-((z-0.65)/0.3)^2] (Fig3) is prescribed shape, not derived from
Eq7-9. Evaluated at DESI DR2 z: 0.295->0.739%, 0.51->2.412%, 0.706->2.8973%, 0.934->1.2244%,
1.321->0.02016%, 1.484->0.00132%, 2.33->~0. No bin at 0.8 or 1.5. Conditional LambdaCDM baseline
Omega_m=0.30 fit gives A_hat=0.02014+-0.01792 (ratio) and 0.02151 [0.00391,0.03943] original-space, chi2
4.905/6 for A=0.03 vs 5.866/6 for A=0. Not evidence for TAL; baseline changes amplitude.
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5. Lab, QED, EEG claims (scope)

Force algebra checked: C(0)=6, C(inf)=1, dC/drho=-(5/rho_c)exp(-rho/rho_c)<0, examples rho=1e-24,1e-23,5e-23,1e-22 => C=5.52419,2.83940,1.03369,1.000227 matching p.1 rounded. Force multiplier ansatz remains unless derived from field dynamics. QED derivation R=(1+r)/(1+2r), r=1/(1+epsilon), 'Jordi Sec V' not in attached PDF, not verifiable here. Lab clock Xi=1e-4 dimensionless, rho_psi=Xi rho_c Phi_t <=1e-68 kg/m3 negligible vs rho_c; mechanism not verified. EEG/HRV r=0.4 p<0.01 predicted with placeholders but without cohort, observable, Phi_t range, N, analysis -- not derived. Antikorrelation C(rho_eff) conditional on fixed baryonic assumptions; r=-0.6 not derived from equation alone.

6. Reproduction and data availability (v16)

To reproduce SPARC check: download Rotmod_LTG.zip, read NGC3198_rotmod.dat line 30.08 kpc, compute v_bar^2=Vgas^2+0.5 Vdisk^2+0.7 Vbul^2, rho_print=v_bar^2*1e6/(G r_m hz,m), C=1+5 exp(-rho/1e-23). For v16 photometric use SBdisk and M_HI from SPARC_Lelli2016c.mrt 28,259 bytes SHA256 5aa0501f6b0d881fa579030e315e7b5b6ef561a5bd3a07472f9929c7e5728243 and Eq5-6. Files for Zenodo v16: Time-Attraction_v16.0.pdf (this file), sparc_175.csv, hi_class.ini, reproduce_NGC3198_v16.py, bao_template_table.csv from DESI audit (DESI mean/cov SHA256: mean 9ac154ab583ce759c0f7eef3c978c7c70a6ead2d18774caceadf1a350a640585 472 bytes, cov 252a143274c8a07c78694c119617d36594f6d7965d00319ca611c6ffb886e509 2547 bytes). Concept DOI stays 10.5281/zenodo.22983681, new version DOI assigned on upload. No v15 PDF file existed for DOI 23220318, only v14.1 PDF 227,612 bytes MD5 b52329276af805802a34f83e17a1f7a7; v16 replaces with explicit name. Boundary: None of these Newtonian tests establishes or rules out relativistic action, metric, photon propagation, lensing; those require separately stated field and matter couplings.

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Appendix A: Numerical examples reproduced from audits

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Constant background rho_b=0.25 rho_c: q=0.25(1+r/L0), at r=1 kpc Fp/(A/kpc^2)=-4.03265,
-dU0/dr=-4.79082, -dUr/dr=-4.03265.
Spherical exponential rho_b=0.3 rho_c exp(-r/2kpc): at r=1 kpc q=0.41681, C=4.29573, rC'=-0.30728,
point-source v_p/v_N=2.07261 vs v0/v_N=2.14546.
Disk rho_b=0.3 rho_c exp(-R/2)exp{-[sqrt(z^2+eps^2)-eps]/0.3}, eps=0.03 kpc, meridional loop r1=1->2,
theta1=0.15, theta2=0.8 rad: Wp/A=+0.405424 kpc^-1, U0 grad contribution -0.405424 (cancellation 2e-8).
Midplane at R=1 kpc: v_p/v_N=2.06667 vs v0/v_N=2.16871 (point source, not extended disk). For centered
spherical, v_p^2=GM C(r)/r, v0^2=GM[C-rC']/r, Omega^2=GM C/r^3, kappa^2=GM[C+rC']/r^3, if C+rC'>0,
apsidal advance 2pi(Omega/kappa-1), at r=1 exponential example kappa^2/Omega^2=0.928469.

v16.0 FINAL -- 2026-10-08 -- All equations dimensionally checked, pair-symmetric, conservative
completion Ur defined, SPARC photometric proxy declared as assumption, redshift prescription unified to
path integral Eq8-9.
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